The Chinese University of Hong Kong, Shenzhen
SDS · School of Data Science
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MAT 3007 — Optimization

Exercise Sheet 6

Exercise E6.1 ((7+7+6=20pts): Stationary Points):
Consider the function

$$f(x) = x_1^3 + x_2^3 - 18x_1^2 - 18x_2^2 - 84x_1$$

Use the first-order necessary condtion (FONC), second order necessary condition (SONC) and second order sufficient condition (SOSC) to find (i) saddle points, (ii) strict local minimizers and(iii) strict local maximizers.

a) saddle points;

b) strict local minimizers;

c) strict local maximizers.

Exercise E6.2 ((8+8+4=20pts): Equality Constrained Problem):
Given a symmetric positive definite matrix $A \in R^{n\times n}$, consider the following optimization problem:

$$\begin{aligned} \text{minimize} \quad & x^T A x \\ \text{subject to} \quad & x^T x = 2 \\ & x_1 = 1 \end{aligned}$$

(where x_1 denotes the first component of vector x)

a) Give the KKT conditions for this problem.

b) When $A = \text{diag}(a_1, \ldots, a_n)$ with distinct diagonal entries, find all KKT points and their total number. (Hint: From the stationarity condition, for $i \geq 2$, either $x_i = 0$ or $a_i = \lambda$)

c) Under the conditions in (b), find the optimal value of this problem.

Exercise E6.3 ((20pts): KKT Conditions):
Construct the KKT conditions for the following linear program (You do not need to solve the problem or find KKT points):

$$\begin{aligned} \text{maximize} \quad & 4x_1 + 3x_2 + 6x_3 \\ \text{s. t.} \quad & 5x_1 + 2x_2 + 3x_3 \leq 15 \\ & 3x_1 + 4x_2 + x_3 \leq 9 \\ & x_1 + x_2 + x_3 \geq 0 \end{aligned}$$

Exercise E6.4 ((20pts): KKT Conditions):
Construct the KKT conditions for the following nonlinear program (You do not need to solve the problem or find KKT points):

$$\begin{aligned} \text{minimize} \quad & x_1 e^{x_1} + (x_2 - 1)^2 + x_3^2 \\ \text{subject to} \quad & x_1 + x_2 \leq 4, \\ & x_3 - x_2^2 \geq 2, \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

Exercise E6.5 ((20pts): 'Quadratic Growth Condition' (QGC) is equivalent to SOSC):
Let $f : X \mapsto \mathbb{R}$ be twice continuously differentiable on $\mathbb{R}^n$. Prove the following 'Quadratic Growth Condition' (QGC) is **equivalent** to SOSC:

QGC: There exists $\alpha, \epsilon > 0$ such that $f(x) - f(x^*) \geq \frac{\alpha}{2}||x - x^*||^2$ for all x in the open ball $B_\epsilon(x^*)$.

SOSC: The point x^* satisfies: (1).$\nabla f(x^*) = 0$; (2).$\nabla^2 f(x^*)$ is a positive definite matrix.

Hint:

a) Prove both **SOSC** $\Rightarrow$ **QGC** and **QGC** $\Rightarrow$ **SOSC**.

b) Review the proof of SOSC in slides, which use second order Taylor expansion.

c) QGC directly implies that x^* is a local minimizer.

d) The open ball $B_\epsilon(x^*)$ is the set of all points whose Euclidean distance to x^* is less than ϵ.

Sheet 6 is due on **April, 13th**. Submit your solutions before **April, 13th**, **11:00pm**.